English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

# Probability and Statistics, Computer Science
Test 2, 9 June 2023: solutions

*English transcription of the supplied handwritten solution notes.*

## Problem 1: leading digit 1

For any distribution of $X$, let $A$ be the probability that the first nonzero decimal digit is 1. Sum the probabilities over $[10^k, 2\cdot 10^k)$, $k \in \mathbb{Z}$, including the corresponding negative intervals for negative values. Since accuracy 0.01 is requested, only a few terms may be needed. For Benford's law, the probability is $\log_{10} 2 \approx 30.1\%$.

### 1. Standard normal

By symmetry, double the sum on the positive half-line. The normal table gives

$$\begin{aligned}\mathbb{P}(1 \le X < 2) &= \Phi(2) - \Phi(1) \approx 0.97725 - 0.84134 = 0.13591,\\ \mathbb{P}(0.1 \le X < 0.2) &\approx 0.57926 - 0.53983 = 0.03943,\\ \mathbb{P}(0.01 \le X < 0.02) &\approx 0.50798 - 0.50399 = 0.00399.\end{aligned}$$

The smaller intervals have total probability bounded above by

$$\mathbb{P}(0 \le X < 0.002) < \Phi(0.01) - \Phi(0) = 0.00399.$$

(In fact $\Phi(0.002) - \Phi(0) \approx 0.0008$, but the first bound is available from the table.) The upper tail is bounded by

$$\mathbb{P}(X \ge 10) \le \mathbb{P}(X \ge 3.99) \approx 0.00003,$$

using the last table entry. The positive contribution is approximately 0.17933 with error below 0.00402. Thus the answer is about 0.35866, with maximum error 0.00802, giving about 36% to accuracy 0.01. The source remarks that 37% would also be accepted under its rough bound, and leaves the answer to accuracy $10^{-10}$ blank.

### 2. Exponential

For $X \sim Exp(1)$, $F_X(t) = 1 - e^{-t}$ for $t \ge 0$. The notes give

$$\mathbb{P}(X \le 0.0002) \approx 0.00020, \qquad \mathbb{P}(X > 10) \approx 0.00005.$$

They sum the central intervals and display

$$A \approx 0.00010 + 0.00985 + 0.08610 + 0.23254 = 0.32859 \approx 33\%.$$

*Transcription note: the source labels the first retained interval* $[0.001, 0.002)$ *but prints its contribution as* $0.00010$*; the displayed approximation is retained as written.*

### 3. Ratio of uniforms

Set $Y = U_1/U_2$ and $Z = U_2$. The inverse map is $U_1 = YZ$, $U_2 = Z$, with Jacobian $z$. Therefore

$$f_{Y,Z}(y,z) = z\mathbb{1}_{\{0\le yz\le 1,\ 0\le z\le 1\}} = z\mathbb{1}_{\{y\ge 0,\ 0\le z\le \min(1,1/y)\}}.$$

Hence

$$f_Y(y) = \int_0^{\min(1,1/y)} z\,dz = \begin{cases} 1/2, & 0 \le y \le 1,\\ 1/(2y^2), & y > 1,\\ 0, & y < 0.\end{cases}$$

The source sketch is flat at height $1/2$ up to 1, then decreases as $1/(2y^2)$. For the tails it gives $\mathbb{P}(Y < 0.00002) = 0.00001$ and, for $t \ge 1$,

$$\mathbb{P}(Y > t) = \int_t^\infty \frac{1}{2y^2}\,dy = \frac{1}{2t}, \qquad \mathbb{P}(Y > 20000) \approx 0.00003.$$

Summing the intervening intervals gives approximately $0.333325 \approx 33\%$. In this case the sum can also be evaluated exactly:

$$\begin{aligned} A &= \frac{1}{2}\sum_{k=1}^{\infty} 10^{-k} + \sum_{k=0}^{\infty}\left(\frac{1}{2\cdot 10^k} - \frac{1}{4\cdot 10^k}\right) \\ &= \frac{1}{2}\frac{1/10}{1-1/10} + \frac{1}{4}\frac{1}{1-1/10} = \frac{1}{3}. \end{aligned}$$

### 4. Powers of two

The notes propose observing and proving that exactly one power of two with $k$ decimal digits begins with 1, for each $k$. Thus there are $k$ such powers up to $10^k$. The number of powers of two below $10^k$ is determined by $2^m < 10^k$, or $m < k\log_2 10$. Their resulting bounds on the proportion are

$$\frac{k}{k\log_2 10 + 1} \le \frac{k}{\lceil k\log_2 10\rceil} \le \frac{k}{k\log_2 10},$$

and both outer terms tend to $1/\log_2 10 = \log_{10} 2$, the Benford probability. *Transcription note: the stated "exactly one" counting claim is reproduced from the handwritten argument; no replacement proof has been supplied.*

## Problem 2: gamma ratio

Set $Y = X_1/(X_1+X_2)$, $Z = X_1 + X_2$. Then $X_1 = YZ$, $X_2 = Z - YZ$, and

$$J = \begin{vmatrix} z & y \\ -z & 1-y \end{vmatrix} = z.$$

The notes start from

$$f_{Y,Z}(y,z) = 16(yz)^2 e^{-2yz}(z-yz)^2 e^{-2(z-yz)} z \mathbb{1}_{\{yz>0, z-yz>0\}},$$

then simplify it in the handwriting to

$$16y^2(1-y)^2 z^3 e^{-2z}\mathbb{1}_{\{z>0, 0<y<1\}}.$$

*Transcription note: this simplification has $z^3$ in the source, although the preceding product contains five powers of $z$. The subsequent workings below follow that displayed exponent.*

The notes write

$$f_Y(y) = 16y^2(1-y)^2\int_0^{\infty} z^3 e^{-2z}\,dz$$

and give three ways to evaluate the constant:

1. Integration by parts gives $\int_0^{\infty} z^3 e^{-2z}\,dz = 3/8$.
2. With $t = 2z$, the integral is $\frac{1}{16}\int_0^{\infty} t^3 e^{-t} dt = \frac{1}{16}\mathbb{E}X^3$ for $X \sim Exp(1)$. Its moment generating function is

$$M_X(\lambda) = \int_0^{\infty} e^{\lambda x} e^{-x} dx = \frac{1}{1-\lambda},$$

   so $M_X'(0) = 1$, $M_X''(0) = 2$, $M_X'''(0) = 6$, again giving $3/8$.
3. Since the expression must be a density on $(0,1)$, integrate $y^2(1-y)^2$ on $(0,1)$ to identify the normalising constant.

For part 2, $Cor(X_1, X_2) = 0$ because the variables are independent by assumption. The notes also conclude

$$Cor\left(\frac{X_1}{X_1+X_2}, X_1+X_2\right) = 0,$$

because the ratio and sum are independent: the displayed joint density factors into a function of $y$ and a function of $z$, including separate support indicators. This factorisation gives $f_{Y,Z} = f_Y f_Z$.

# Problem 3: CLT approximation

1. "Standard CLT calculation."

2. For $X_i \sim Ber(1/1000)$,

$$\mu = 1/1000, \qquad \sigma = \sqrt{999/10^6} \approx 10^{-1.5}.$$

The handwritten expression for the third moment is

$$\rho = \frac{1}{1000}\left(1 - \frac{1}{1000}\right)^3 + \frac{999}{1000}\left(0 - \frac{1}{1000}\right)^3 \approx 10^{-3}.$$

It uses the Berry–Esseen error bound $\rho/(2\sigma^3\sqrt{n})$. For $n = 100$, the displayed estimate is

$$\frac{\rho}{2\sigma^3\sqrt{100}} \approx 10^{-3} \cdot \frac{1}{2} \cdot 10^{4.5} \cdot \frac{1}{10} \approx 1.58.$$

For desired errors 0.001, 0.01 and 0.1, the handwritten conclusions are respectively

| error | bound on $\sqrt{n}$ | displayed bound on $n$ |
|---|---|---|
| 0.001 | $10^{1.5} \cdot 10^3/2$ | $10^9/2$ |
| 0.01 | $10^2 \cdot 10^{1.5}/2$ | $10^7/2$ |
| 0.1 | $10 \cdot 10^{1.5}/2$ | $10^5/2$ |

*The source writes the bounds with denominator 2 after squaring; those figures are transcribed unchanged.*

For comparison, when $p = 1/2$, the notes use $\mu = 1/2$, $\sigma = 1/2$, $\rho = 1/8$. The corresponding requirements are

| error | $\sqrt{n}$ | $n$ |
|---|---|---|
| 0.001 | $> 500$ | $> 250000$ |
| 0.01 | $> 50$ | $> 2500$ |
| 0.1 | $> 5$ | $> 25$ |

# Problem 4

"Standard calculation."